

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter-2022

Date: 09/01/2023

Course: B. Pharmacy

Subject Name: Novel Drug Delivery System

Max Marks: 75

Sem: VII

Subject Code: BP704T

Duration: 3Hrs

Instructions –

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What are ideal characters of polymers?
- ii) Give types of ocular inserts.
- iii) Write factors affecting formulation of Controlled release drug delivery systems.
- iv) Enlist advantages of implantable drug delivery system.
- v) Write the methods of microencapsulation.
- vi) Name the Marketed products of Transdermal Patches.
- vii) Give the types of medicated IUD's.
- viii) Write the different approaches for gastric retention.
- ix) Give examples of chemical penetration enhancers.
- x) Give the disadvantages of copper intrauterine devices.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What is Gastro retentive Drug Delivery System? Give its advantages & disadvantages with approaches for GRDDS.
- ii) Define Transdermal drug delivery system (TDDS)? Give its advantages and disadvantages. Describe permeation enhancer with examples.
- iii) What is Nasal DDS? Give its advantages & disadvantages with formulation of Nasal Sprays.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Biopharmaceutic Characteristics of Drug to qualify for CDDS.
- ii) Explain the Theories of Mucoadhesion.
- iii) Write the challenges in delivering drug to the eye.
- iv) Explain about Microballoons as gastro adhesive drug delivery system.
- v) Explain concept, advantages and disadvantages of liposomes.
- vi) State various methods to prepare nanoparticles.
- vii) Describe the formulations of nasal sprays.
- viii) Describe intrauterine drug delivery system with applications
- ix) Write a note on Non-Effervescent Systems for FDDS.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

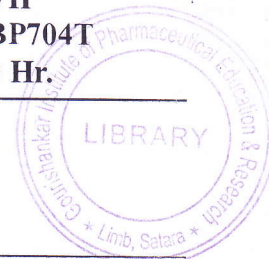
Supplementary Examination Summer - 2023

Course : B. Pharmacy
Subject Name : Novel drug delivery System
Max Marks : 75

Date: 23/06/2023
Sem: VII
Subject Code: BP704T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks



Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Enlist factors affecting formulation of Controlled drug delivery system.
- ii) Outline Classification of Polymers.
- iii) Give applications of Microencapsulation.
- iv) Give Disadvantages of Implantable drug delivery system.
- v) Enlist Excipients used in Nasal spray formulations.
- vi) Give role of Propellants in Aerosol along with their examples.
- vii) Give examples of Carriers used in targeted drug delivery.
- viii) What are Monoclonal Antibodies? Give any 2 applications of the same.
- ix) Describe role of Viscosity enhancers in improving Ocular Bioavailability
- x) Describe advantages of Intra-uterine devices.

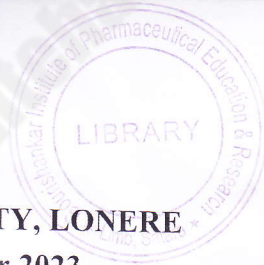
Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What is GRDDS? Give the various approaches for Gastro-retention.
- ii) Define TDDS? Give its advantages & disadvantages. Describe permeation enhancer with examples.
- iii) Explain in detail the various types of Implants.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain the Various theories of Mucoadhesion.
- ii) Explain Air suspension and Multi-orifice centrifugal process of Microencapsulation.
- iii) Explain the Basic components of transdermal drug delivery system.
- iv) Describe formulation of Nasal sprays
- v) Write a note on Dry Powder Inhalers.
- vi) Write a note on Monoclonal antibodies and their applications.
- vii) Write a note on various Ocular inserts.
- viii) Write a note on various Intra-uterine devices
- ix) Describe in detail the various applications of Polymers.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Supplementary/Regular Winter 2023

Date: 09/01/2024

Course: B. Pharmacy
Subject Name: Novel Drug Delivery System
Max Marks: 75

Sem: VII
Subject Code: BP704T BP404T
Duration: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1 Solve the following.

(10x2 =20M)

1. Write factors affecting the formulation of Controlled release drug delivery systems
2. Define microsphere and microcapsules.
3. What are the stages in mucoadhesion?
4. Write the excipients in nasal spray formulations.
5. Write types of contact lenses.
6. Enlist applications of intrauterine drug delivery
7. What are the types of niosomes?
8. Write the criteria followed to select polymers for Controlled release drug delivery systems.
9. Write note on ALZET osmotic pump?
10. State basic components of transdermal drug delivery.

Q.2 Solve any two of the following

(2x10 = 20M)

1. Explain different formulation approaches of Transdermal drug delivery system.
2. Describe various theories of mucoadhesion with their significance in designing mucoadhesive products.
3. What are liposomes? Explain their characterization methods and applications in detail.

Q.3 Solve any seven of the following

(7x5 = 35M)

1. Classify the polymers used to modify the drug release.
2. Describe Ion Exchange Resins based controlled release formulation.
3. Write in detail about the microencapsulated drug delivery systems.
4. Describe the effervescent gastroadhesive drug delivery system.
5. Define and classify the different microparticulate drug delivery systems.
6. State various methods to prepare nanoparticles.
7. Write mechanisms of controlled drug release in ophthalmic drug delivery.
8. What are the advantages and disadvantages of intrauterine drug delivery systems?
9. What is a pulmonary route of administration? Explain in brief about drug powder inhalers.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: VII

Subject Name: NOVEL DRUG DELIVERY SYSTEM

Subject Code: BP704T

Max Marks: 75

Date: 20-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10 X 2) = 20
i)	Describe nanoparticles along with their general properties.	
ii)	Explain ideal properties of bioadhesive polymer	
iii)	What factors affect the designing of modified drug delivery system?	
iv)	What is targeted drug delivery? Give its applications.	
v)	Write note on conservation method of microencapsulation.	
vi)	Define and compare active & passive targeting.	
vii)	What are the disadvantages of conventional ocular drug delivery system	
viii)	What is mucoadhesive drug delivery system? Give its application.	
ix)	Classify Liposome according to structure?	
x)	What are ion exchange resins?	
Q.2.	Long Answers (Answer 2 out of 3)	(10 X 2) = 20
i)	Explain in detail different methods for formulation of TDDS along with evaluation.	
ii)	Explain in detail various methods of preparation of liposomes	
iii)	Explain in detail formulation methods for nanoparticles along with advantages of nanoparticulate delivery.	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Explain the different theories of mucoadhesion.	
ii)	Explain the evaluation parameters for transdermal patches.	
iii)	Write a short note on contact- lens.	
iv)	What are advantages & disadvantages of implantable drug delivery system	
v)	Explain metered dose inhaler.	
vi)	Describe vapour pressure activated implantable device.	
vii)	Write a short note on biodegradable polymers.	
viii)	Explain the different barriers in ocular drug delivery	
ix)	Describe DSC & TGA studies of evaluation of polymer.	